Jiaan Guo | Jack Guo

Data Science & Applied ML

New York metropolitan area jackguo2005@gmail.com github.com/Jack311045

EDUCATION

New York University Expected May 2027

Bachelor of Arts, joint major in Data Science and Mathematics · GPA 3.64/4.00 through Spring 2026

Relevant coursework: Data Structures, Data Management and Analysis, Fundamentals of Machine Learning, Causal Inference, Mathematical Statistics, Linear and Nonlinear Optimization (in progress)

EXPERIENCE

Sustentra Jun-Sep 2026 · Remote

Data Science & Engineering Intern

- Contributed to Python document workflows connecting S3, Textract OCR, and OpenAI extraction with schema-backed JSON and field-level source context.
- Translated review requirements into versioned schemas, validation checks, and traceable evidence flows with engineering and ESG/audit collaborators.
- Helped build a seven-stage Streamlit auditor prototype; some stages use prepared fixtures rather than connected live services.

GigaCloud Technology Jul-Aug 2025 · Suzhou, China

Data Science Intern

- Analyzed 100,000+ shipment records to study package-dimension differences, billing changes, and surcharge exposure.
- Built geometry and carrier-threshold features and compared UPS/FedEx measurement stages; shared packaging and risk-identification recommendations.

CITIC Securities Jun-Jul 2025 · Henan, China

Marketing / Data Analyst Intern

- Analyzed options-trading and private-fund datasets for financial-product research, product positioning, and internal strategy discussions.

Veterans Coaching Program / Project Jan-May 2026 · Remote

Research & Data Support

- Researched grant opportunities and organized structured trackers supporting fundraising operations and follow-up.

SELECTED PROJECTS

ChessLens 2026-Present · Active development

- Built streaming Lichess PGN ingestion with deterministic IDs, HMAC player identifiers, sharded Parquet, resumable manifests, and staged validation.
- Recorded ingestion report: 10.64M accepted games, 718.56M moves, 43 shards. Developed DuckDB/dbt analytics and leakage-aware classical policy, value, and ranking baselines.

Recipe Discovery & Semantic Search Jan-May 2026

- Combined SBERT and TF-IDF retrieval with SigLIP image/text search, dietary filters, negative feedback, and rating-aware reranking.
- Implemented PyTorch K-means with k-means++ initialization, Lloyd updates, empty-cluster reseeding, and multiple restarts.

Met Museum Cross-Modal Search Dec 2025-Jan 2026

- Built DINOv2/text-feature retrieval with contrastive projection heads and FAISS for text-to-image and image-to-text search.

SKILLS

Languages: Python, SQL

Data: pandas, NumPy, DuckDB, dbt, PyArrow/Parquet, SciPy, statsmodels, ETL, feature engineering, statistical analysis

ML & retrieval: PyTorch, scikit-learn, LightGBM, Sentence Transformers, TF-IDF, SigLIP, DINOv2, FAISS

Engineering & document systems: FastAPI, Pydantic, JSON Schema, AWS S3, Textract, OpenAI API, pytest, Hypothesis, Git/GitHub Actions, Docker, Ruff, mypy, Streamlit

Development workflow: AI-assisted development

Recorded project figures reflect saved reports; they are not represented as newly rerun results.